Companion LXXXVI: Neural Joint Likelihoods for RDCC — Combining Transport, Minkowski Functionals, and Shear Power in a Unified Flow-Based Model

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Abstract

Weak-lensing surveys provide multiple complementary probes of the large-scale structure, including topological persistence diagrams, Minkowski functionals, and the shear power spectrum. Previous Companions developed transport-based statistics, multi-probe Fisher forecasts, and cross-correlation analyses for the Relaxation-Driven Cyclic Cosmology (RDCC). In this work, we introduce a unified neural joint likelihood that combines these observables into a single flow-based generative model. The likelihood is constructed using normalizing flows trained on multi-probe summary vectors, enabling differentiable inference of the relaxation parameter α. This framework captures non-Gaussian topology, morphological structure, and two-point correlations simultaneously, yielding a powerful and flexible tool for RDCC parameter estimation in Euclid-like surveys.

1 Introduction

Weak-lensing convergence maps contain rich information about the large-scale structure. Different observables probe different aspects of this structure:

- transport-based persistence-diagram statistics capture non-Gaussian topological information,
- Minkowski functionals quantify morphological features,
- the shear power spectrum $C_\ell^{\gamma\gamma}$ captures two-point correlations.

Previous Companions developed each of these probes individually and constructed pairwise joint analyses. The present Companion unifies all three into a single neural joint likelihood, enabling coherent and differentiable inference of the RDCC relaxation parameter α.

2 Multi-Probe Summary Vector

We define the combined summary vector

$$\mathbf{s} = \{\mathcal{T}_\ell, V_0(\nu), V_1(\nu), V_2(\nu), C_\ell^{\gamma\gamma}\},$$

where:

- $\mathcal{T}_\ell$ are transport-based topological statistics,
- $V_i(\nu)$ are Minkowski functionals,
- $C_\ell^{\gamma\gamma}$ is the shear power spectrum.

This vector captures topology, morphology, and two-point structure simultaneously.

3 Flow-Based Likelihood Model

We construct a normalizing flow f_θ such that

$$\mathbf{z} = f_\theta(\mathbf{s})$$

maps the summary vector to a Gaussian latent space. The likelihood is

$$\mathcal{L}(\mathbf{s} \mid \alpha) = \mathcal{N}(f_\theta(\mathbf{s}); \mu(\alpha), \Sigma(\alpha)) \left| \det \frac{\partial f_\theta}{\partial \mathbf{s}} \right|.$$

The flow is trained on simulations across a grid of α values.

4 Differentiable RDCC Inference

Because the likelihood is differentiable with respect to α, we can perform:

- gradient-based maximum-likelihood estimation,
- Hamiltonian Monte Carlo,
- variational inference,
- neural posterior estimation.

The derivative

$$\frac{\partial \mathcal{L}}{\partial \alpha}$$

is computed via automatic differentiation through the flow.

5 Advantages of the Unified Approach

The neural joint likelihood provides:

- a coherent combination of topology, morphology, and two-point structure,
- improved constraints on α through multi-probe synergy,
- a flexible generative model for summary statistics,
- robustness to non-Gaussianity and survey noise,
- differentiable inference pipelines for Euclid-like surveys.

This framework generalizes classical Gaussian likelihoods and captures non-linear, non-Gaussian RDCC signatures.

6 Conclusion

We introduced a unified neural joint likelihood that combines topological transport statistics, Minkowski functionals, and the shear power spectrum into a single flow-based generative model. This framework captures complementary information from topology, morphology, and two-point correlations, enabling powerful and differentiable inference of the RDCC relaxation parameter α. The present Companion completes the transition from geometric theory to state-of-the-art multi-probe likelihood modeling for Euclid-like surveys.

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